LANGUAGES FOR AI, MODULE 2
DATE 02/07/2026

Time: 2 hours.

You should have three sheets of paper.
Write your name and identification number in the **upper right corner** of ALL the sheets, so that it appears at least once per sheet (not per-page).
Write the answers to exercises 1-2, 3-4 and 5-6 on three different sheets of paper.

You are allowed to use only the following connectives: $\wedge, \vee, \neg, \rightarrow$. **Any other connective will be considered invalid.**

1. (5 points) Consider the language of propositional logic. Use natural deduction to prove that the following holds, or find a counter-example to show that it does not hold (remember that $\neg F$ is only a shorthand for $F \rightarrow \bot$).

   - $\vdash ((A \rightarrow B) \wedge (A \rightarrow \neg B)) \rightarrow \neg A$
   - $(A \rightarrow B) \vdash (B \rightarrow C) \rightarrow (A \rightarrow C)$

   For students of academic year 2024/25 only: verify for each of the previous formulas whether they are: valid/invalid, satisfiable/unsatisfiable using truth tables. Briefly explain your answer.

2. (5 points) Transform and simplify the following propositional logic formula into an equivalent formula in Disjunctive Normal Form. You are allowed to use only the following connectives: $\wedge, \vee, \neg, \rightarrow$. **Any other connective will be considered invalid.**

$$\big((A \leftrightarrow B) \vee C\big) \rightarrow \big(D \wedge (A \rightarrow C)\big)$$

3. An alarm system is modeled using propositional logic. The propositional variables are defined as follows:

   - A: the alarm is sounding
   - I: there is a real intrusion
   - F: the sensor is faulty
   - R: the alarm is disabled

   The system is described by the following formulas:

   (a) $\neg R \rightarrow (A \rightarrow (I \vee F))$
   (b) $F \rightarrow \neg I$
   (c) $R \rightarrow \neg A$
   (d) $I \rightarrow A$

   If the alarm is sounding, can we be sure that there is an intrusion? Motivate your answer.

4. Consider a first-order language whose domain consists of databases and records. The language contains the following predicates:

   - Consistent(d): database d is consistent;
   - Stores(d,r): database d stores record r;
   - Corrupt(r): record r is corrupt.

   Assume the following first-order axioms:

(a) **Consistency excludes corruption**

$$\forall d \forall r (Consistent(d) \wedge Stores(d, r) \rightarrow \neg Corrupt(r))$$

(b) **Existence of a consistent database**

$$\exists d \, Consistent(d)$$

(c) **Complete storage by consistent databases**

$$\forall r \exists d (Consistent(d) \wedge Stores(d, r))$$

(d) **Existence of a corrupt record**

$$\exists r \, Corrupt(r)$$

Questions: 1) Is the theory given by the five axioms above satisfiable in standard first-order logic? Motivate your answer.

2) is the sentence

$$\forall d \forall e (Consistent(d) \wedge Consistent(e) \rightarrow d = e)$$

a logical consequence of the theory given by the five axioms ?

5. (6 points) A logistics company manages deliveries across multiple warehouses. Each warehouse is represented as `warehouse(Name, Deliveries)`, where `Deliveries` is a list whose elements can be a successful delivery (e.g., `delivery(Client, Amount)`, where `Amount` is the value in euros) or a failed delivery attempt (e.g., `failed`). Write a recursive Prolog predicate `successful_deliveries(Warehouses, Report)` that scans the `Warehouses` list (each with its own inner list of deliveries) and constructs a new list `Report` containing `Client-Amount` pairs for every successful delivery with an amount strictly greater than 50 euros. Failed deliveries and deliveries with an amount of 50 euros or less must be ignored.

Example Queries:

```
?- successful_deliveries([
     warehouse(milan, [delivery(smith, 120), failed, delivery(jones, 30)]),
     warehouse(turin, [delivery(green, 75), delivery(black, 200)])
   ], L).
L = [smith-120, green-75, black-200].

?- successful_deliveries([warehouse(rome, [failed, delivery(white, 20)])], L).
L = [].
```

Write the Prolog code for the `successful_deliveries/2` predicate below.

6. (6 points) A company manufactures four types of drone kits: Mini (M), Standard (S), Racing (R), and Pro (P). You need to find a valid production configuration based on the following constraints:

   (a) **Quantities:** The number of kits of each type must be between 0 and 40 (inclusive).

   (b) **Total Kits:** The total number of kits must be exactly **80 kits**.

   (c) **Cost Constraint:** The cost per kit is: Mini 3 euro, Standard 6 euro, Racing 12 euro, and Pro 20 euro. The `total value` of the production must be exactly `444 euro`.

   (d) **Proportion Requirement:** The number of Standard kits must be at least three times the sum of Racing and Pro kits combined.

   (e) **Conditional Requirement:** If any Pro kits are produced, then at least 5 Racing kits must also be produced. If no Pro kits are produced, this constraint does not apply.

   (f) **Optimization:** Among all valid configurations, find the one that **maximizes** the number of Pro kits produced.

Write the full Prolog/CLP code defining the predicate `drone_production(M, S, R, P)`, using `library(clpfd)`.